

Question Number	Answer	Additional guidance	Mark
7(a)	A calculation in which: - total population of Santa Cruz in 2020 calculated (1) - total population of Santa Cruz in 2025 calculated (1) - population given to whole number (1)	<u>Example of calculation</u> $(17000 \div 85) \times 100 = 20\,000$ $20\,000 \times 1.064^5 = 27\,273.3 / 27\,273.328 / 27\,273.33$ {27 273 / 27 000} Correct answer with no working scores full marks ecf applies {26400 / 26000} = 2 marks	(3)
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7(b)	positive correlation between the number of recorded non endemic species and the number of tourists increased	ACCEPT number of recorded non endemic species increased as the number of tourists increased	(1)

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7(c)(i)	A calculation in which: • $N(N-1)$ correctly calculated (1) • $\sum n(n-1)$ correctly calculated (1) • correct substitution to obtain D to 1 d.p. (1)	<u>Example of calculation</u> $(147 \times 146) = 21462$ 4684 4.6 (4.58 = 2 marks) <table border="1"> <thead> <tr> <th>Species</th><th>Number of individuals (n)</th><th>(n-1)</th><th>n(n-1)</th></tr> </thead> <tbody> <tr><td>A</td><td>21</td><td>20</td><td>420</td></tr> <tr><td>B</td><td>2</td><td>1</td><td>2</td></tr> <tr><td>C</td><td>4</td><td>3</td><td>12</td></tr> <tr><td>D</td><td>13</td><td>12</td><td>156</td></tr> <tr><td>E</td><td>54</td><td>53</td><td>2862</td></tr> <tr><td>F</td><td>15</td><td>14</td><td>210</td></tr> <tr><td>G</td><td>6</td><td>5</td><td>30</td></tr> <tr><td>H</td><td>32</td><td>31</td><td>992</td></tr> <tr> <td></td><td>Total (N)=147</td><td></td><td>$\sum n(n-1)=4684$</td></tr> </tbody> </table> Correct answer with no working scores full marks ECF applied	Species	Number of individuals (n)	(n-1)	n(n-1)	A	21	20	420	B	2	1	2	C	4	3	12	D	13	12	156	E	54	53	2862	F	15	14	210	G	6	5	30	H	32	31	992		Total (N)=147		$\sum n(n-1)=4684$	(3)
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7(c)(ii)	An answer that includes five of the following points: • reduction in biodiversity (1) • as forest is habitat for many species of plants (1) • (because) populations will decrease because loss of forest will result in {reduced habitat for animals / reduced food} / increased competition between (animal) species} (1) • causing reduction in species richness (1) • (because) loss of forest will result in reduction of genetic diversity (1) • (due to) reproduction in {smaller / isolated} populations (1)	Accept possible increase in species that have blackberry as part of food chain Accept reduction in forest habitat (for many species unqualified) / blackberry {outcompetes (the forest) plants / introduces disease} Accept animals not adapted to feed on blackberry / blackberry is poisonous to the animals Accept descriptions of what animals would compete for e.g. nesting sites Accept reduction in gene pool	(5)